

DBMS Lab: SQL Joins and Transactions

Part A: Create Tables

```
CREATE TABLE roles(role_id INT PRIMARY KEY, role_name VARCHAR(50));
CREATE TABLE users(user_id INT PRIMARY KEY,name VARCHAR(50),email
VARCHAR(50),role_id INT);
CREATE TABLE departments(dept_id INT PRIMARY KEY,dept_name
VARCHAR(50));
CREATE TABLE employees(emp_id INT PRIMARY KEY,emp_name
VARCHAR(50),dept_id INT,manager_id INT,salary DECIMAL(10,2));
CREATE TABLE customers(customer_id INT PRIMARY KEY,customer_name
VARCHAR(50),city VARCHAR(50));
CREATE TABLE orders(order_id INT PRIMARY KEY,customer_id INT,order_date
DATE,total_amount DECIMAL(10,2));
CREATE TABLE accounts(account_no INT PRIMARY KEY,holder_name
VARCHAR(50),balance DECIMAL(10,2));
```

```
mysql> create database dbms_lab;
Query OK, 1 row affected (0.07 sec)

mysql> use dbms_lab;
Database changed
mysql> CREATE TABLE roles(role_id INT PRIMARY KEY, role_name VARCHAR(50));
Query OK, 0 rows affected (0.07 sec)

mysql> CREATE TABLE users(user_id INT PRIMARY KEY,name VARCHAR(50),email VARCHAR(50),role_id INT);
Query OK, 0 rows affected (0.07 sec)

mysql> CREATE TABLE departments(dept_id INT PRIMARY KEY,dept_name VARCHAR(50));
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE employees(emp_id INT PRIMARY KEY,emp_name VARCHAR(50),dept_id INT,manager_id INT,salary DECIMAL(10,2));
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE customers(customer_id INT PRIMARY KEY,customer_name VARCHAR(50),city VARCHAR(50));
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE orders(order_id INT PRIMARY KEY,customer_id INT,order_date DATE,total_amount DECIMAL(10,2));
Query OK, 0 rows affected (0.10 sec)

mysql> CREATE TABLE accounts(account_no INT PRIMARY KEY,holder_name VARCHAR(50),balance DECIMAL(10,2));
Query OK, 0 rows affected (0.07 sec)

mysql> insert into roles(role_id,role_name) values (1,'Rama'),(2,'Sita'),(3,'suresh'),(4,'manu');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> INSERT INTO users VALUES (101, 'Rahul', 'rahul@gmail.com', 1),(102, 'Priya', 'priya@gmail.com', 2),(103, 'Anil', 'anil@gmail.com', 3),(104, 'Sneha', 'sneha@gmail.com', 4);
Query OK, 4 rows affected (0.04 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> insert into departments values(1,'cse'),(2,'ece'),(3,'csit'),(4,'aids');
```

```

mysql> insert into departments values(1,'cse'),(2,'ece'),(3,'csit'),(4,'aids');
Query OK, 4 rows affected (0.04 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> insert into employees(111,'harika',1,202,6000.00),(222,'Karuna',2,204,5000.00),(333,'swetha',3,303,9000.00),(444,'anusha',404,2000.00);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '111,'harika',1,202,6000.00),(222,'Karuna',2,204,5000.00),(333,'swetha',3,303,9000.00),(444,'anusha',404,2000.00);
mysql> insert into employees values(111,'harika',1,202,6000.00),(222,'Karuna',2,204,5000.00),(333,'swetha',3,303,9000.00),(444,'anusha',404,2000.00);
ERROR 1136 (21S01): Column count doesn't match value count at row 4
mysql> insert into employees values(111,'harika',1,202,6000.00),(222,'Karuna',2,204,5000.00),(333,'swetha',3,303,9000.00),(444,'anusha',4,404,2000.00);
Query OK, 4 rows affected (0.04 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> insert into customers values(1,'hema','hyd'),(2,'krishana','bhel'),(3,'divya','kondapur'),(4,'keerthi','delhi');
Query OK, 4 rows affected (0.04 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> insert into orders values(1001,1,'2026-08-12',3000.00),(1002,2,'2026-09-10',7000.00),(1003,3,'2026-03-13',9000.00),(1004,4,'2026-06-24',2000.00);
-> insert into orders values(1001,1,'2026-08-12',3000.00),(1002,2,'2026-09-10',7000.00),(1003,3,'2026-03-13',9000.00),(1004,4,'2026-06-24',2000.00);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'insert into orders values(1001,1,'2026-08-12',3000.00),(1002,2,'2026-09-10',7000.00),(1003,3,'2026-03-13',9000.00),(1004,4,'2026-06-24',2000.00);
mysql> insert into orders values(1001,1,'2026-08-12',3000.00),(1002,2,'2026-09-10',7000.00),(1003,3,'2026-03-13',9000.00),(1004,4,'2026-06-24',2000.00);
Query OK, 4 rows affected (0.04 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> insert into accounts values(1010,'sai',30000.00),(2020,'aryan',4000.00),(3030,'harini',700.00),(4040,'willam',30.00);
Query OK, 4 rows affected (0.16 sec)
Records: 4 Duplicates: 0 Warnings: 0

```

Q1. Display employee names with their department names using INNER JOIN.

SELECT emp_name,dept_name FROM employees e INNER JOIN departments d ON e.dept_id=d.dept_id;

```

mysql> select emp_name,dep_name from employees e inner join departments d on e.dept_id=d.dept_id;
ERROR 1054 (42S22): Unknown column 'dep_name' in 'field list'
mysql> select emp_name,dept_name from employees e inner join departments d on e.dept_id=d.dept_id;
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| harika   | cse       |
| Karuna   | ece       |
| swetha   | csit      |
| anusha   | aids      |
+-----+-----+
4 rows in set (0.00 sec)

mysql> select * from employees;
+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | manager_id | salary |
+-----+-----+-----+-----+-----+
| 111    | harika   | 1       | 202        | 6000.00 |
| 222    | Karuna   | 2       | 204        | 5000.00 |
| 333    | swetha   | 3       | 303        | 9000.00 |
| 444    | anusha   | 4       | 404        | 2000.00 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> select * from departments;
+-----+-----+
| dept_id | dept_name |
+-----+-----+
| 1       | cse       |
| 2       | ece       |
| 3       | csit      |
| 4       | aids      |
+-----+-----+

```

Q2. Display all users with their assigned roles using LEFT JOIN.

SELECT name,email,role_name FROM users u LEFT JOIN roles r ON
u.role_id=r.role_id;

```
mysql> select * from users;
+-----+-----+-----+-----+
| user_id | name  | email           | role_id |
+-----+-----+-----+-----+
| 101     | Rahul | rahul@gmail.com | 1       |
| 102     | Priya | priya@gmail.com | 2       |
| 103     | Anil  | anil@gmail.com  | 3       |
| 104     | Sneha | sneha@gmail.com | 4       |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> select *from roles;
+-----+-----+
| role_id | role_name |
+-----+-----+
| 1       | Rama      |
| 2       | Sita      |
| 3       | suresh    |
| 4       | manu      |
+-----+-----+
4 rows in set (0.00 sec)

mysql> select name,email,role_id from users u left join roles r on u.role_id=r.role_id;
ERROR 1052 (23000): Column 'role_id' in field list is ambiguous
mysql> select u.name,u.email,u.role_id from users u left join roles r on u.role_id=r.role_id;
+-----+-----+-----+
| name  | email           | role_id |
+-----+-----+-----+
| Rahul | rahul@gmail.com | 1       |
| Priya | priya@gmail.com | 2       |
| Anil  | anil@gmail.com  | 3       |
| Sneha | sneha@gmail.com | 4       |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

Q3. Display all roles even if no users are assigned using RIGHT JOIN.

SELECT name,role_name FROM users u RIGHT JOIN roles r ON u.role_id=r.role_id;

```
mysql> SELECT name,role_name FROM users u RIGHT JOIN roles r ON u.role_id=r.role_id;
+-----+-----+
| name  | role_name |
+-----+-----+
| Rahul | Rama      |
| Priya | Sita      |
| Anil  | suresh    |
| Sneha | manu      |
+-----+-----+
4 rows in set (0.01 sec)
```

Q4. Simulate FULL OUTER JOIN in MySQL.

SELECT name,email,role_name FROM users u LEFT JOIN roles r ON
u.role_id=r.role_id UNION SELECT name,email,role_name FROM users u RIGHT
JOIN roles r ON u.role_id=r.role_id;

```
Database changed
mysql> SELECT name,email,role_name FROM users u LEFT JOIN roles r ON u.role_id=r.role_id UNION SELECT name,email,role_name FROM users
u RIGHT JOIN roles r ON u.role_id=r.role_id;
+-----+-----+-----+
| name | email          | role_name |
+-----+-----+-----+
| Rahul | rahul@gmail.com | Rama      |
| Priya | priya@gmail.com | Sita      |
| Anil  | anil@gmail.com  | suresh    |
| Sneha | sneha@gmail.com | manu      |
+-----+-----+-----+
4 rows in set (0.05 sec)
```

Q5. Display employee and manager names using SELF JOIN.

SELECT e.emp_name Employee,m.emp_name Manager FROM employees e LEFT
JOIN employees m ON e.manager_id=m.emp_id;

```
mysql> SELECT e.emp_name Employee,m.emp_name Manager FROM employees e LEFT JOIN employees m ON e.manager_id=m.emp_id;
+-----+-----+
| Employee | Manager |
+-----+-----+
| harika   | NULL    |
| Karuna   | NULL    |
| swetha   | NULL    |
| anusha   | NULL    |
+-----+-----+
4 rows in set (0.01 sec)
```

Q6. Display Order ID, Customer Name, City and Total Amount.

SELECT o.order_id,customer_name,city,total_amount FROM customers c JOIN orders o
ON c.customer_id=o.customer_id;

```
mysql> SELECT o.order_id,customer_name,city,total_amount FROM customers c JOIN orders o ON c.customer_id=o.customer_id;
+-----+-----+-----+-----+
| order_id | customer_name | city   | total_amount |
+-----+-----+-----+-----+
| 1001     | hema          | hyd    | 3000.00      |
| 1002     | krishana      | bhel   | 7000.00      |
| 1003     | divya         | kondapur | 9000.00      |
| 1004     | keerthi       | delhi  | 2000.00      |
+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

Q7. Find department-wise employee count, average salary and maximum salary.

SELECT dept_name,COUNT(*),AVG(salary),MAX(salary) FROM employees e JOIN departments d ON e.dept_id=d.dept_id GROUP BY dept_name;

```
mysql> SELECT dept_name,COUNT(*),AVG(salary),MAX(salary) FROM employees e JOIN departments d ON e.dept_id=d.dept_id GROUP BY dept_name;
+-----+-----+-----+-----+
| dept_name | COUNT(*) | AVG(salary) | MAX(salary) |
+-----+-----+-----+-----+
| cse       | 1        | 6000.000000 | 6000.00     |
| ece       | 1        | 5000.000000 | 5000.00     |
| csit      | 1        | 9000.000000 | 9000.00     |
| aids      | 1        | 2000.000000 | 2000.00     |
+-----+-----+-----+-----+
4 rows in set (0.05 sec)
```

Q8. Display employees earning >50000 in CSE.

SELECT emp_name,salary,dept_name FROM employees e JOIN departments d ON e.dept_id=d.dept_id WHERE salary>50000 AND dept_name='CSE';

```
mysql> SELECT emp_name,salary,dept_name FROM employees e JOIN departments d ON e.dept_id=d.dept_id WHERE salary>50000 AND dept_name='cse';
Empty set (0.00 sec)
```

Q9. Display customer-wise total purchase in descending order.

SELECT customer_name,SUM(total_amount) TotalPurchase FROM customers c JOIN orders o ON c.customer_id=o.customer_id GROUP BY customer_name ORDER BY TotalPurchase DESC;

```
mysql> SELECT customer_name,SUM(total_amount) TotalPurchase FROM customers c JOIN orders o ON c.customer_id=o.customer_id GROUP BY customer_name ORDER BY TotalPurchase DESC;
+-----+-----+
| customer_name | TotalPurchase |
+-----+-----+
| divya         | 9000.00       |
| krishana      | 7000.00       |
| hema          | 3000.00       |
| keerthi       | 2000.00       |
+-----+-----+
4 rows in set (0.00 sec)
```

Q10. Create Students, Courses, Faculty, Enrollments and display Student, Course, Faculty using joins.

SELECT student_name,course_name,faculty_name FROM students s JOIN enrollments e ON s.student_id=e.student_id JOIN courses c ON e.course_id=c.course_id JOIN faculty f ON c.faculty_id=f.faculty_id;

```
mysql> create table faculty( faculty_id int primary key, faculty_name varchar(50));
Query OK, 0 rows affected (0.21 sec)

mysql> CREATE TABLE courses ( course_id INT PRIMARY KEY,course_name VARCHAR(50),faculty_id INT);
Query OK, 0 rows affected (0.05 sec)

mysql> create table students(student_id int primary key, student_name varchar(50));
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE enrollments ( student_id INT,course_id INT);
Query OK, 0 rows affected (0.19 sec)

mysql> INSERT INTO faculty VALUES (1, 'Dr. Kumar'),(2, 'Dr. Reddy'),(3, 'Dr. Sharma'),(4, 'Dr. Rao');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> INSERT INTO courses VALUES(101, 'DBMS', 1),(102, 'Java', 2),(103, 'Python', 3),(104, 'OS', 4);
Query OK, 4 rows affected (0.04 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> INSERT INTO students VALUES(1,'Rahul'),(2,'Priya'),(3,'Anil'),(4,'Sneha');
Query OK, 4 rows affected (0.04 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> insert into enrollments values (1,101),(2,102),(3,103),(4,104);
Query OK, 4 rows affected (0.05 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> select *from students;
```

student_id	student_name
1	Rahul
2	Priya
3	Anil
4	Sneha

```
4 rows in set (0.00 sec)
```

```
mysql> select *from faculty;
```

faculty_id	faculty_name
1	Dr. Kumar
2	Dr. Reddy
3	Dr. Sharma
4	Dr. Rao

```
4 rows in set (0.00 sec)
```

```
mysql> select *from course;
```

ERROR 1146 (42S02): Table 'dbms_lab.course' doesn't exist

```
mysql> select *from courses;
```

course_id	course_name	faculty_id
101	DBMS	1
102	Java	2
103	Python	3
104	OS	4

```
4 rows in set (0.00 sec)
```

```
mysql> select *from enrollments;
```

student_id	course_id
1	101
2	102
3	103
4	104

```
mysql> SELECT student_name,course_name,faculty_name FROM students s JOIN enrollments e ON s.student_id=e.student_id JOIN courses c ON
e.course_id=c.course_id JOIN faculty f ON c.faculty_id=f.faculty_id;
```

student_name	course_name	faculty_name
Rahul	DBMS	Dr. Kumar
Priya	Java	Dr. Reddy
Anil	Python	Dr. Sharma
Sneha	OS	Dr. Rao

```
4 rows in set (0.00 sec)
```

Part B: Transactions

Q11. Insert a new account and COMMIT.

START TRANSACTION; INSERT INTO accounts VALUES(104,'Anil',25000);
COMMIT;

```
mysql> START TRANSACTION; INSERT INTO accounts VALUES(104,'Anil',25000); COMMIT;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> select * from accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
| 104 | Anil | 25000.00 |
| 1010 | sai | 30000.00 |
| 2020 | aryan | 4000.00 |
| 3030 | harini | 700.00 |
| 4040 | willam | 30.00 |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

Q12. Insert a record and ROLLBACK.

START TRANSACTION; INSERT INTO accounts VALUES(105,'Ravi',10000);
ROLLBACK;

```
mysql> START TRANSACTION; INSERT INTO accounts VALUES(106,'Ravi',10000); ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.00 sec)

Query OK, 0 rows affected (0.04 sec)

mysql> select * from accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
| 104 | Anil | 25000.00 |
| 1010 | sai | 30000.00 |
| 2020 | aryan | 4000.00 |
| 3030 | harini | 700.00 |
| 4040 | willam | 30.00 |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

Q13. Transfer ₹5000 from Account 101 to 102.

START TRANSACTION; UPDATE accounts SET balance=balance-5000 WHERE
account_no=101; UPDATE accounts SET balance=balance+5000 WHERE
account_no=102; COMMIT;


```
mysql> select * from accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
| 104 | Anil | 25000.00 |
| 1010 | sai | 30000.00 |
| 2020 | aryan | 4000.00 |
| 3030 | harini | 700.00 |
| 4040 | willam | 30.00 |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> START TRANSACTION; UPDATE accounts SET balance=balance-5000 WHERE account_no=1010; UPDATE accounts SET balance=balance+5000 WHERE account_no=2020; COMMIT;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.04 sec)
Rows matched: 1 Changed: 1 Warnings: 0

Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

Query OK, 0 rows affected (0.00 sec)

mysql> select * from accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
| 104 | Anil | 25000.00 |
| 1010 | sai | 25000.00 |
| 2020 | aryan | 9000.00 |
| 3030 | harini | 700.00 |
| 4040 | willam | 30.00 |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

Q14. Simulate failed transaction and rollback.

START TRANSACTION; UPDATE accounts SET balance=balance-3000 WHERE account_no=101; UPDATE accounts SET balance=balance+3000 WHERE account_no=999; ROLLBACK;

```
mysql> START TRANSACTION; UPDATE accounts SET balance=balance-3000 WHERE account_no=1010; UPDATE accounts SET balance=balance+3000 WHERE account_no=999; ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

Query OK, 0 rows affected (0.04 sec)

mysql> select * from accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
| 104 | Anil | 25000.00 |
| 1010 | sai | 25000.00 |
| 2020 | aryan | 9000.00 |
| 3030 | harini | 700.00 |
| 4040 | willam | 30.00 |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

Q15. Demonstrate SAVEPOINT.

START TRANSACTION; UPDATE accounts SET balance=balance+2000 WHERE account_no=101; SAVEPOINT s1; UPDATE accounts SET balance=balance-1000 WHERE account_no=101; ROLLBACK TO s1; COMMIT;

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE accounts SET balance=balance+2000 WHERE account_no=1010;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SAVEPOINT s1;
Query OK, 0 rows affected (0.04 sec)

mysql> UPDATE accounts SET balance=balance-1000 WHERE account_no=101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql> ROLLBACK TO s1;
Query OK, 0 rows affected (0.00 sec)

mysql> commit;
Query OK, 0 rows affected (0.04 sec)

mysql> select * from accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
| 104 | Anil | 25000.00 |
| 1010 | sai | 27000.00 |
| 2020 | aryan | 9000.00 |
| 3030 | harini | 700.00 |
| 4040 | willam | 30.00 |
+-----+-----+-----+
5 rows in set (0.00 sec)
```